

JavaScript Comments & Variables

Recap of the Day 2 videos: writing comments that explain intent, and declaring variables with `let` and `const` — the containers every test you ever write will use.

Comments

Comments are notes for humans — JavaScript ignores them. Use them to explain **why** code exists, not to repeat what it does.

```
// single-line comment
/* multi-line comment:
   useful for longer explanations
   or temporarily disabling code */
```

Variables: `let`, `const` (and `var`)

- **`const`** — the default choice. The variable cannot be reassigned.
- **`let`** — use when the value needs to change (counters, accumulators).
- **`var`** — the legacy keyword. Avoid it; you will see why in Day 3 (scoping).

```
const courseName = "Playwright"; // fixed value
let score = 10;                  // will change
score = score + 5;                // OK
// courseName = "Cypress";       // Error! const cannot be reassigned
```

Naming rules & conventions

- Names may contain letters, digits, `_` and `$` — but cannot **start** with a digit.
- Names are case-sensitive: `userName` and `username` are different variables.
- Reserved words (`let`, `if`, `class`...) cannot be used as names.
- Convention: **camelCase** for variables — `firstName`, `totalPrice`, `isLoggedIn`.

Quick reference

Keyword	Reassignable?	When to use
<code>const</code>	No	Default — values that never change
<code>let</code>	Yes	Values that change (counters, loops)
<code>var</code>	Yes	Legacy — avoid in new code

Practice exercises

Do these in VS Code, run them with Node, and bring questions to the next live Q&A.

1. Declare `const courseName = "Playwright"`, then try to reassign it. Read the error message carefully — you will meet it again.

Hint: `TypeError: Assignment to constant variable.`

2. Declare `let score = 10`, add 5 to it, and print the result.
3. Write a small script with a one-line comment on top describing what it does, and one multi-line comment explaining a decision.
4. Which of these names are invalid, and why? `1name`, `first-name`, `let`, `$total`, `user_2`.

Hint: Two are invalid, one is a reserved word — and two are perfectly fine.

5. Create `a = 1` and `b = 2` (with `let`), then swap their values using a third variable. Print both.

Bonus challenge — Create three variables — `name`, `age`, `city` — and print them in a single `console.log` as one sentence. (After Day 4 you can rewrite it with a template literal.)